

Legrand

GRMS Platform MQTT Interface Protocol (2019-03-08 Update)

Guest Control Platform MQTT Interface Protocol (Local Area Network)

Change List

Summary	Content Description	Date & Time
Modified subscription/publication topics, device status query, and device list acquisition	1. All subscription/publication topics/SMARTHOTEL/GENERAL/SYNCDATETIME remain SMARTHOTEL/GERNERAL/SYNCDATETIME 2. Added <code>deviceType</code> (device type) to device status query 3. Modified the device list acquisition protocol: changed <code>RoomType</code> to <code>RoomTypeNo</code>	2019-3-8 16:39
Modified device list acquisition	Changed dynamic room type code to fixed key "RoomTypeNo"; modified <code>hotelCode</code> to <code>hotelId</code> and <code>deviceCode</code> to <code>deviceId</code>	2019-3-6 16:46
Modified device list	Changed the room number in the device list	2019-2-29 11:22

acquisition	to room type code	
Added in-room service status	Added a protocol to implement in-room service status control for smart devices such as mobile phones, speakers, or TVs to control in-room service status on demand. RCU system suppliers shall monitor relevant protocols, including: • In-room service status control;	2019-2-22 17:42
Rule definition	All subscription/publication topics are in uppercase; keys are all in lowercase; values are case-insensitive	2019-2-22 17:42

Protocol Overview

Item	Details								
Communication Protocol	Implemented based on MQTT (Apache Apollo)								
QOS	2								
Identification Information	Use Topic to distinguish different types of information. Uniformly define identification information according to the source, destination, level, etc., of guest control information for easy sending/subscription. Classification & Subpath/Topic: <table border="1"> <tr> <td>Classification</td><td>Topic</td></tr> <tr> <td>Classification</td><td>STATUS</td></tr> <tr> <td>Classification</td><td>COMMAND</td></tr> <tr> <td>Classification</td><td>GERNERAL</td></tr> </table>	Classification	Topic	Classification	STATUS	Classification	COMMAND	Classification	GERNERAL
Classification	Topic								
Classification	STATUS								
Classification	COMMAND								
Classification	GERNERAL								

	<table><tr><td>Classification</td><td>RCU</td></tr><tr><td>Classification</td><td>SERVICE</td></tr><tr><td>Classification</td><td>PMS</td></tr><tr><td>Classification</td><td>DOORLOCK</td></tr><tr><td>Classification</td><td>TV</td></tr><tr><td>Classification</td><td>SPEAKER</td></tr><tr><td>Classification</td><td>PUBLIC AREA</td></tr></table>	Classification	RCU	Classification	SERVICE	Classification	PMS	Classification	DOORLOCK	Classification	TV	Classification	SPEAKER	Classification	PUBLIC AREA
Classification	RCU														
Classification	SERVICE														
Classification	PMS														
Classification	DOORLOCK														
Classification	TV														
Classification	SPEAKER														
Classification	PUBLIC AREA														
Payload Format	JSON														
Communication Security	SSL 3.0														
Communication Objects	- Internal systems - Subscription to guest control cloud service "SMARTHOTEL/STATUS/[hotelId]"; - Subscription to guest control monitoring "SMARTHOTEL"; ----- - All manufacturers - Subscription to "SMARTHOTEL/GENERAL"; - Subscription to "SMARTHOTEL/COMMAND/[hotelId]/PMS"; ----- - Intelligent guest control service - Subscription to "SMARTHOTEL/COMMAND/														
Synchronization Strategy	The status of in-room devices or guest rooms shall be based on feedback from guest control hardware devices, not control status. For example: the status of a reading light refers to the status of the reading light control module (or the actual status of the reading light if the hardware has a detection circuit function).														

Notes:

1. Status Codes: See precautions for details; supplementary information will be provided if needed.

Value	Description
0200	Success
0301	Response data is empty
0408	Request timeout
0302	Incorrect request parameters
0500	Internal program error

2. Guest Control Device Classification

Device Name	Device Type Code	Related Status
Door Lock	DoorLock	Door lock open status, door opening records (opening method, opener/door card ID, opening time, etc.);
Light Switch	Light	On/Off;
Dimmer Switch	DimmerLight	On/Off, Brightness;
Curtain	Curtain	Open, Close, Pause, Position;
Air Conditioner	AC	On/Off, Fan Speed, Working Mode, Set Temperature, Current Temperature;
TV	TV	On/Off (Standby), Channel Change, Volume, Menu Operation;

Speaker	Speaker	Play, Stop, Pause, TTS Playback;
Power Supply	Power	On/Off
Gateway Device	Host	N/A
Motion Sensor	MotionSensor	Occupied/Unoccupied
Door Sensor	DoorSensor	Door Open/Closed
Socket	Outlet	On/Off
Water Leak Detector	WaterSensor	Water Leak/No Leak
Gas Detector	GasSensor	Gas Leak/No Leak
Smoke Detector	SmokeSensor	Smoke Detected/None
Acoustooptic Alarm	AcoustoopticSensor	Alarm Triggered/None
Safe Sensor	SafeSensor	Safe Opened/Closed
Infrared Transmitter	Infrared	(Refer to TV)
Scene Mode	Scene	(Mode List) Pre-comfort, Welcome, Sleep, Wake-up, Energy-saving, Occupied Energy-saving, Guest-defined, Unoccupied, Patrol, etc.;
Bathroom Exhaust Fan	ExhaustFan	On/Off
Window Sensor	WindowSensor	Window Open/Closed
Guide Board	Guideboard	Guide light On/Off for designated guest rooms

3. In-room Service Classification

Service Name	Service Code	Related Status
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Do Not Disturb	DND	Enabled/Disabled
Make Up Room	MUR	Enabled/Disabled
Emergency Button	SOS	Triggered/Not Triggered
Dining in Room	DIR (Dining in Room)	Enabled/Disabled
Green Hotel	GH (Green Hotel)	Enabled/Disabled
Tray Collection	CTT (Collect the Tray)	Enabled/Disabled
Laundry Service	LS (Laundry Service)	Enabled/Disabled
Laundry Return	LR (Laundry Return)	Enabled/Disabled
Butler Service	BS (Butler Service)	Enabled/Disabled

4. Definition of Return Values

Item	Details
Return Value	[Optional] Asynchronous communication; no meaningless return values such as "Received" or "Processed" shall be transmitted. Instead, the status shall be returned.
Path	/SMARTHOTEL/STATUS/{hotelId}/XXX
Parameters	Parameter Name
	msgId
	status
	info
	message

	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "status": "0200", "info": "success", "message": "success", "timestamp": 1544515904 }</pre>

[GNL] Synchronize Time

Description	For clock synchronization of all platform devices
Path	SMARTHOTEL/GERNERAL/SYNCDATETIME
Parameters	Parameter Name
	msgId
	current
	timeZone
	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "current": "2019-01-18 12:00:00", "timeZone": "+8", "timestamp": 1547117456 }</pre>

	}
--	---

[PMS] Guest Check-in

Description	When a guest checks in, the guest control platform forwards check-in related information through this topic.			
Path	SMARTHOTEL/COMMAND/[hotelId]/PMS/CHECKIN			
Parameters	Parameter Name			
	RCU MQTT Protocol	RCU MQTT Protocol	RCU MQTT Protocol	RCU MQTT Protocol
	msgId	Y	String	The unique identifier of the message, in the format: supplier identification number_hotel number_self-incrementing serial number, where the self-incrementing serial number is used uint64.
	hotelId	Y	string	Hotel number
	businessKey	Y	string	Business order number, uniquely identifying the guest staying overnight
	roomNo	Y	string	It consists of the building number, floor number, room number, and sub-room number;
	floorNo	N	string	The floor number where the guest room is located;
	shares	Y	int	Identify check-in type: 0: single occupancy; 1: multiple occupancy, need to split the bill; 2: multiple people staying together;
	lastName	N	string	Guest's last name, which may be in Chinese, English, or other languages;
	title	N	string	Guest appellations include not only gender and job titles, etc. A value of 0 indicates that the information is incomplete.

	<table><tr><td>vip</td><td>N</td><td>string</td><td>VIP notes;</td></tr><tr><td>checkin</td><td>N</td><td>string</td><td>Check-in time (YYYY-MM-DD HH:MM:SS)</td></tr><tr><td>latestCheckout</td><td>N</td><td>string</td><td>Latest check-out time (YYYY-MM-DD HH:MM:SS)</td></tr><tr><td>timestamp</td><td>Y</td><td>int</td><td>Current timestamp (in seconds)</td></tr></table>	vip	N	string	VIP notes;	checkin	N	string	Check-in time (YYYY-MM-DD HH:MM:SS)	latestCheckout	N	string	Latest check-out time (YYYY-MM-DD HH:MM:SS)	timestamp	Y	int	Current timestamp (in seconds)
vip	N	string	VIP notes;														
checkin	N	string	Check-in time (YYYY-MM-DD HH:MM:SS)														
latestCheckout	N	string	Latest check-out time (YYYY-MM-DD HH:MM:SS)														
timestamp	Y	int	Current timestamp (in seconds)														
Example	<pre>{ "msgId": "0021_8000110_1024", "hotelId": "8000110", "businessKey": "C80001100001882181", "roomNo": "A_5F_8529_2", "floorNo": "5", "shares": 0, "lastName": "Wang", "title": "Dr. ", "vip": "Gold Member", "checkin": "2019-01-17 16:21:37", "latestCheckout": "2019-01-18 12:00:00", "latestCheckout": "2019-01-18 12:00:00", "timestamp": 1547117456 }</pre>																

[PMS] Guest Check-out

Description	The guest control platform forwards check-out related information through this topic.
Path	SMARTHOTEL/COMMAND/[hotelId]/PMS/CHECKOUT
Parameters	Parameter Name
	msgId

	hotelId
	roomNo
	businessKey
	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "hotelId": "8000110", "businessKey": "C80001100001882181", "roomNo": "A_5F_8529_2", "timestamp": 1547117456 }</pre>

[PMS] Guest Room Change

Description	The guest control platform forwards room change related information through this topic.
Path	SMARTHOTEL/COMMAND/[hotelId]/PMS/CHANGEROOM
Parameters	Parameter Name
	msgId
	hotelId
	businessKey
	changefrom
	roomNo
	floorNo
	shares

	checkin
	latestCheckout
	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "hotelId": "8000110", "businessKey": "C80001100001882181", "changeFrom": "A_5F_8529_2", "roomNo": "A_6F_8629_2", "floorNo": "6", "shares": 0, "checkin": "2019-01-17 16:21:37", "latestCheckout": "2019-01-18 12:00:00", "timestamp": 1547117456 }</pre>

[PMS] Guest Extension of Stay

Description	The guest control platform forwards stay extension related information through this topic.
Path	SMARTHOTEL/COMMAND/[hotelId]/PMS/EXTENSIONRESERVATION
Parameters	Parameter Name
	msgId
	hotelId
	businessKey
	roomNo
	shares

	checkin
	latestCheckout
	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "hotelId": "8000110", "businessKey": "C80001100001882181", "roomNo": "A_6F_8629_2", "shares": 0, "checkin": "2019-01-17 16:21:37", "latestCheckout": "2019-01-18 12:00:00", "timestamp": 1547117456 }</pre>

[RCU] In-room Device Control

Description	Control devices in RCU rooms through hotel ID, room number, device ID, and device values.
Path	SMARTHOTEL/COMMAND/[hotelId]/RCU/DEVICE CONTROL
Parameters	Parameter Name
	msgId
	hotelId
	roomNo
	controlDevice
	deviceId
	deviceType

	sourceType
	cmdDetails
	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "hotelId": "8000110", "roomNo": "A_6F_8629_2", "controlDevice": [{ "deviceId": "jsdkgj492 ", "deviceType": "AC", "sourceType": "1", "cmdDetails": { "targetTemp": "19", "mode": "cooling", "vent": "high", "switch": "1" }] }, "timestamp": 1547117456 }</pre>

Remarks:

1. Commands and Parameters

Applicable Device	Key (string)	Value (string)
Door Lock	unlock	1-unlock (RCU shall not control door locks in principle)
Light	switch	1-on, 0-off

	flip (Toggle: switch on/off)	1
Dimmer Light	switch	1-on, 0-off
	brightness	60 (0~100, set to 0 when switch == 0)
Curtain	control	0-pause, 1-open, 2-close
	position	0~100 (%)
Air Conditioner	switch	1-on, 0-off
	temp	19 (16~33)
	currentTemp	XX (MIN~MAX)
	mode (Mode)	Cooling (auto, cooling, heating, or dehumidification)
	vent (Fan Speed)	High (auto, low, middle, high)
	adjustVent (Fan Speed Adjustment)	(2: increase by 2 levels, -1: decrease by 1 level)
	adjustTemp (Temperature Adjustment)	(5: increase by 5 degrees, -1: decrease by 1 degree)
TV	switch	0-Standby (TV can only standby by default; power on/off requires TV support), 1-on, 2-off
	Mute	1
	Vol_Up (Volume Up)	1
	Vol_Down (Volume Down)	1

	Channel_Up (Channel Up)	1
	Channel_Down (Channel Down)	1
	Back (Back)	1
	Menu	1
	Sign (Signal Switch)	1
	Up	1
	Down	1
	Left	1
	Right	1
	OK (Confirm)	1
	Homepage	1
	Airplay (Screen Mirroring)	1
Speaker	switch	N/A
	Play	0-stop, 1-play
	Pause	1
	Vol_Up (Volume Up)	1
	Vol_Down (Volume Down)	1
	Song_Next (Next Song)	1
	Song_Pre (Previous Song)	1
	Mute	1

Socket	switch	1-on, 0-off
Gateway	switch	1-on, 0-off
Scene Mode	switch	1

[RCU] Query Device Status

Description	Query the status of in-room devices through hotel ID, room number, and device ID.
Path	SMARTHOTEL/COMMAND/[hotelId]/RCU/QUERYSTATUS
Parameters	Parameter Name
	msgId
	hotelId
	roomNo
	deviceId
	deviceType
	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "hotelId": "8000110", "roomNo": "01050100", "deviceId": "dhgfkja1", "deviceType": "DimmerLight", "timestamp": 1547117456 }</pre> Query the status of all devices in the room when both <code>deviceType</code> and <code>deviceId</code> are empty.

Return Value	Refer to "Status Synchronization".
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[RCU] In-room Device Status Synchronization

Description	The RCU system publishes the status of specified devices in designated guest rooms, which can be a single device, a group of devices (including sensors), or the status of in-room scenes.
Path	SMARTHOTEL/STATUS/[hotelId]/RCU
Parameters	Parameter Name
	msgId
	hotelId
	roomNo
	status
	deviceId
	deviceType
	online
	sourceType
	sourceMsg
	settingValue
	currentValue
	timestamp
Example	{ "msgId": "0021_8000110_1024", "hotelId": "8000110",

	<pre>"roomNo":"A_6F_8629_2", "status":[{ "deviceId":"dhgfkja1", "deviceType":"DimmerLight", "online":"1", "sourceType":"1", "sourceMsg":"0021_8000110_1028", "settingValue":{ "switch":"1", "brightness":"75" }, "currentValue":{ "switch":"1", "brightness":"75" } }, { "deviceId":"oujri8s3", "deviceType":"AC", "online":"1", "sourceType":"1", "sourceMsg":"0021_8000110_1028", "settingValue":{ "switch":"1", "temp":"19", "mode":"cooling", "vent":"high" }, "currentValue":{ "switch":"1", "temp":"19", "mode":"cooling", "vent":"high",</pre>
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	<pre>"currentTemp":"22" } }, { "deviceId":"rs4dl31", "deviceType":"DoorSensor" "online":"1", "sourceType":"1", "sourceMsg":"0021_8000110_1028", "currentValue":{ "switch":"1" } }}, "timestamp": 1547117456 }</pre>
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[RCU] Query Service Status

Description	Query the status of in-room services through hotel ID and room number.
Path	SMARTHOTEL/COMMAND/[hotelId]/RCU/QUERYSERVICE
Parameters	Parameter Name
	msgId
	hotelId
	roomNo
	serviceId
	timestamp
Example	<pre>{ "msgId":"0021_8000110_1024",</pre>

	<pre>"hotelId": "8000110", "roomNo": "A_6F_8629_2", "serviceId": "dnd", "timestamp": 1547117456 }</pre>
Return Value	Refer to "Status Synchronization".

[RCU] In-room Service Status Control

Description	Publish in-room service status control commands to the RCU system
Path	SMARTHOTEL/COMMAND/[hotelId]/RCU/ROOMSERVICECONTROL
Parameters	Parameter Name
	msgId
	hotelId
	roomNo
	status
	serviceId
	value
	sourceType
	sourceMsg
	timestamp
Example	<pre>{ "msgId": "0021_8000110_1024", "hotelId": "8000110",</pre>

	<pre>"roomNo": "A_6F_8629_2", "status": { "serviceld": "dnd", "value": "0", "sourceType": "SPEAKER", "sourceMsg": "0021_8000110_1028" }, "timestamp": 1547117456 }</pre>
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[RCU] In-room Service Status Synchronization

Description	The RCU system publishes the service status of designated guest rooms
Path	SMARTHOTEL/STATUS/[hotelId]/ROOMSERVICE
Parameters	Parameter Name
	msgId
	hotelId
	roomNo
	status
	serviceld
	value
	sourceType
	sourceMsg
	timestamp
Example	{

```
"msgId":"0021_8000110_1024",
"hotelId": "8000110",
"roomNo":"A_6F_8629_2",
"status":[{
  "servicId":"dnd",
  "value":"0",
  "sourceType":"0",
  "sourceMsg":"0021_8000110_1028"
}, {
  "servicId":"mur",
  "value":"1",
  "sourceType":"0",
  "sourceMsg":"0021_8000110_1028"
}, {
  "servicId":"SOS",
  "value":"0",
  "sourceType":"0",
  "sourceMsg":"0021_8000110_1028"
}, {
  "servicId":"dinningInRoom",
  "value":"0",
  "sourceType":"0",
  "sourceMsg":"0021_8000110_1028"
}, {
  "servicId":"greenHotel",
  "value":"0",
  "sourceType":"0",
  "sourceMsg":"0021_8000110_1028"
}],
"timestamp": 1547117456
}
```

[RCU] Request Hotel Device List

Description	Obtain room type, room, and device list information through hotel ID.
Path	SMARTHOTEL/COMMAND/[hotelId]/RCU/DEVICELIST
Parameters	Parameter Name
	msgId
	hotelId
	timestamp
Example	{ "msgId": "0021_8000110_1024", "hotelId": "8000110", "timestamp": 1547117456 }
Return Value	Refer to "Obtain Device List".

[RCU] Obtain Device List

Description	Obtain room type, room, and device list information through hotel ID.
Path	SMARTHOTEL/STATUS/[hotelId]/RCU/DEVICELIST

Return Value:

Parameter	Parameter Name	Type	Mandatory	Description
data	Response Data	Object	Yes	{ RoomTypeList [

				RoomTypeNo
				RoomTypeNa me
				RoomList [RoomNo]
				DeviceList [HotelId [Mandatory]
				RoomTypeNo [Mandatory]
				DeviceTypeC ode [Mandatory]
				DeviceTypeN ame [Optional]
				OrderType [Optional]
				DisplayType [Optional] DeviceId [Mandatory]
				DeviceName [Optional]
				DeviceImage [Optional] AreaId [Optional]
				AreaName

				[Optional]
				DeviceAlias [Optional]
				IrChip [Optional] (Mandatory for infrared devices)
				DeviceStatus [Optional]
				]
				]
				}

Return Data:

```
{
  "roomTypeList": [
    {
      "roomTypeNo": "1001",
      "roomTypeName": "Standard Room",
      "roomList": [
        "A_1F_101_2",
        "A_2F_201_2"
      ],
      "deviceList": [
        {
          "hotelId": "1188",
          "roomTypeNo": "1001",
          "deviceTypeCode": "lighting",
          "displayType": "lighting",
          "deviceTypeName": "Lighting",
          "deviceId": "1",
          "deviceName": "Corridor Light",
```

```
"orderType": "SC",
"statusInfo": {
  "val": "0"
},
"relationImage": "",
"reqFun": "0",
"alias": "",
"areald": 1,
"areaName": "Bedroom"
},
{
  "hotelId": "1188",
  "roomTypeNo": "1001",
  "deviceTypeCode": "scene",
  "displayType": "scene",
  "deviceTypeName": "Scene",
  "deviceId": "10",
  "deviceName": "Welcome Mode",
  "orderType": "SC",
  "statusInfo": {
    "val": "0"
  },
  "relationImage": "http://sh.jobosz.com:6060/jobobs/static/image/visitor.png",
  "areald": null,
  "areaName": ""
},
{
  "hotelId": "1188",
  "roomTypeNo": "1001",
  "deviceTypeCode": "scene",
  "displayType": "scene",
  "deviceTypeName": "Scene",
  "deviceId": "14",
```

```
"deviceName": "TV Mode",
"orderType": "SC",
"statusInfo": {
  "val": "0"
},
"areald": null,
"areaName": "",
},
{
  "hotelId": "1188",
  "roomTypeNo": "1001",
  "deviceTypeCode": "ac",
  "displayType": "ac",
  "deviceTypeName": "Air Conditioner",
  "deviceId": "1",
  "deviceName": "Air Conditioner",
  "orderType": "AC",
  "statusInfo": {
    "mode": "1",
    "setTemper": "26",
    "temper": "28",
    "fanSpeed": "1"
  },
  "areald": null,
  "areaName": ""
},
{
  "hotelId": "1188",
  "roomTypeNo": "1001",
  "deviceTypeCode": "curtain",
  "displayType": "curtain",
  "deviceTypeName": "Curtain",
  "deviceId": "1",
```

```
"deviceName": "Curtain",
"orderType": "SC",
"statusInfo": {
  "val": "0"
},
"relationImage": "",
"cnSwitchList": [
  {
    "switchGoupId": 1,
    "switchid": "21",
    "switchStatus": "1",
    "switchStatusName": "On"
  },
  {
    "switchGoupId": 1,
    "switchid": "22",
    "switchStatus": "0",
    "switchStatusName": "Off"
  }
],
"areald": null,
"areaName": ""
},
{
  "hotelId": "1188",
  "roomTypeNo": "1001",
  "deviceTypeCode": "curtain",
  "displayType": "curtain",
  "deviceTypeName": "Curtain",
  "deviceId": "2",
  "deviceName": "Curtain 1",
  "orderType": "SC",
  "statusInfo": {
```

```
    "val": "0"
  },
  "relationImage": "",
  "cnSwitchList": [
    {
      "switchGoupld": 2,
      "switchid": "23",
      "switchStatus": "1",
      "switchStatusName": "On"
    },
    {
      "switchGoupld": 2,
      "switchid": "24",
      "switchStatus": "0",
      "switchStatusName": "Off"
    }
  ],
  "areald": null,
  "areaName": ""
},
{
  "hotelld": "1188",
  "roomTypeNo": "1001",
  "deviceTypeCode": "curtain",
  "displayType": "curtain",
  "deviceTypeName": "Curtain",
  "deviceId": "4",
  "deviceName": "Curtain 33",
  "orderType": "SC",
  "statusInfo": {
    "val": "0"
  },
  "relationImage": "",
```

```
"cnSwitchList": [  
  {  
    "switchGoupId": 4,  
    "switchid": "27",  
    "switchStatus": "1",  
    "switchStatusName": "On"  
  },  
  {  
    "switchGoupId": 4,  
    "switchid": "28",  
    "switchStatus": "0",  
    "switchStatusName": "Off"  
  }  
],  
"areald": null,  
"areaName": ""  
},  
{  
  "hotelId": "1188",  
  "roomTypeNo": "1001",  
  "deviceTypeCode": "curtain",  
  "displayType": "curtain",  
  "deviceTypeName": "Curtain",  
  "deviceId": "3",  
  "deviceName": "Curtain 2",  
  "orderType": "SC",  
  "statusInfo": {  
    "val": "0"  
  },  
  "relationImage": "",  
  "cnSwitchList": [  
    {  
      "switchGoupId": 3,
```

```
"switchid": "29",  
"switchStatus": "1",  
"switchStatusName": "On"  
},  
{  
  "switchGoupId"
```